Lab-4

1. Objectives

A. Working with objects and classes.
B. Implement class methods that modify object state

2. Submission

Once your program works correctly and the output format is correct, submit your work using the following Git commands:

```
git add <file_name>

git commit -m "Commit message"

git push
```

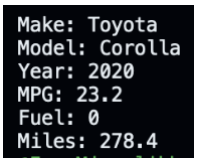
4. Instructions

Continue working with the Car class we implemented in class. Do the following:

1) Add private double data field named mileage.
2) Add private double data field named fuel_capacity representing the fuel tank capacity of the car.
3) Add private double data field named fuel_level that stores the current fuel level in the car.
4) Implement the accessor (get) function for fuel_level property.
5) Update the user-defined constructor so that it accepts five arguments. The constructor should initialize mileage and fuel_level properties with zeros.

```
Car(string make_, string model_, int year_, double
MPG_, double fuel_capacity_);
```

6) Modify printInfo method and print car's information in the following format:



7) Implement class method refuel to add fuel to the car. The method should display how much fuel was added and fuel lever after refueling. If the amount of fuel exceeds the tank capacity, display the excess amount.

```
void Car::refuel(double gallons);
```

```
Refueling...
Fuel added: 2 gallons
Fuel level: 7 gallons
```

```
Refueling...
Fuel added: 5.0431 gallons
Excess fuel: 11.9569 gallons
Fuel level: 12 gallons
```

Implement class method drive which simulates car movement for the given distance. If the given distance exceeds the distance the car can cover with the current fuel_level, the method should display covered distance and how many miles left. The method should also display the remaining fuel_level.

```
void Car::drive(double distance);
```

```
Distance covered: 123 miles (0 miles left)
Fuel level: 6.65517 gallons
```

```
Distance covered: 154.4 miles (345.6 miles left)
Fuel level: 0 gallons
```

In the main.cpp file create a Car object named toyota with the following specifications.

- Make: Toyota
- Model: Corolla
- Year: 2020
- MPG: 23.2
- Fuel capacity: 12 gallons

Run the provided code. (g++ main.cpp Car.cpp)

5. Sample Run

Refueling...

Fuel added: 5 gallons

Fuel level: 5 gallons

Enter gallons: 7

Enter distance: 125

Refueling...

Fuel added: 7 gallons

Fuel level: 12 gallons

Distance covered: 125 miles (0 miles left)

Fuel level: 6.61207 gallons

Make: Toyota

Model: Corolla

Year: 2020

MPG: 23.2

Fuel: 6.61207

Miles: 125

Enter gallons: 0

Enter distance: 98

Refueling...

Fuel added: 0 gallons

Fuel level: 6.61207 gallons

Distance covered: 98 miles (0 miles left)

Fuel level: 2.38793 gallons

Make: Toyota

Model: Corolla

Year: 2020

MPG: 23.2

Fuel: 2.38793

Miles: 223

Enter gallons: 1

Enter distance: 200

Refueling...

Fuel added: 1 gallons

Fuel level: 3.38793 gallons

Distance covered: 78.6 miles (121.4 miles left)

Fuel level: 0 gallons

Make: Toyota

Model: Corolla

Year: 2020

MPG: 23.2

Fuel: 0

Miles: 301.6